

Reproducing Human Selective-Listening Trends with a Pretrained Feature-Gain Model

A qualitative replication on a custom Common Voice evaluation set

Griffith, Hess & McDermott (2026, Nature Human Behaviour)

Group meeting · Single pretrained checkpoint · 15–20 min

Can a gain-based model explain selective listening?

The human problem

At a noisy party we can follow **one voice** and ignore others.

The model (Griffith et al. 2026)

Applies a learned **feature gain** that boosts features matching a cue voice and suppresses the rest.

The claim

One optimized gain mechanism predicts BOTH successes and failures of human listening across many conditions.

Our task

Feed the **authors' pretrained checkpoint** our own Common Voice test set and check whether the **behavioral trends** reproduce.

The task, per trial

1. **Cue** — a sample of the target voice
2. **Mixture** — target + distractor(s)
3. **Report** the target's middle word

(from an 800-word vocabulary)

Parameters are frozen: this tests behavior at inference, not retraining.

Building the evaluation set — one shared pipeline



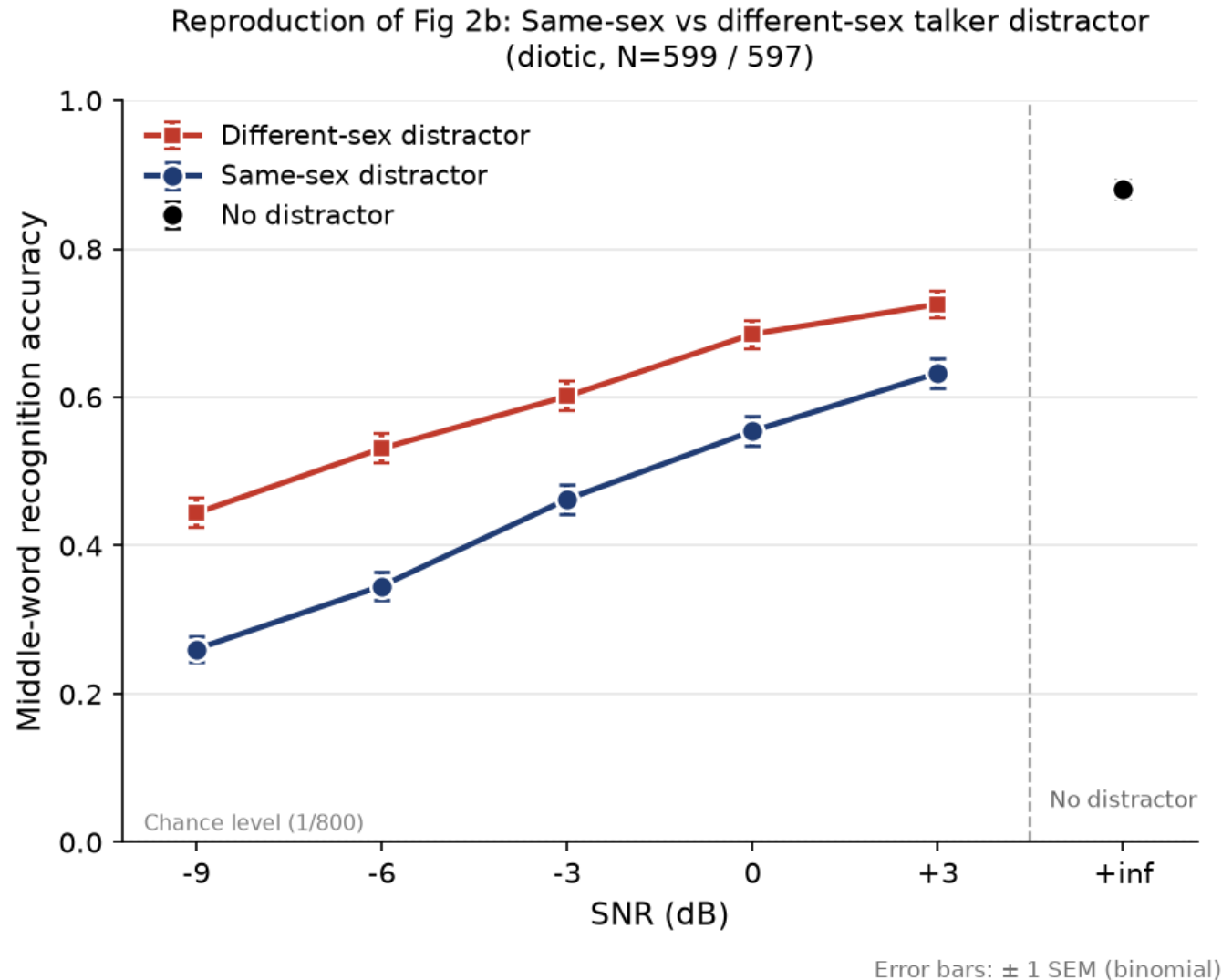
Anchor = the paper's rule: pick the target word first, then cut a slice centered on it

- Structural constraints: cue & target = same speaker / different recordings; distractor = different speaker; gender-balanced targets.
- Paper-compatible scoring: a prediction is correct if it matches any in-vocabulary word in the target excerpt.
- Model validity check: 88.6% clean-speech accuracy; +inf lands at 88.0% (chance = 0.1%).

PART 1

Distractor sex · a single competing talker

Accuracy rises with SNR; different-sex is easier



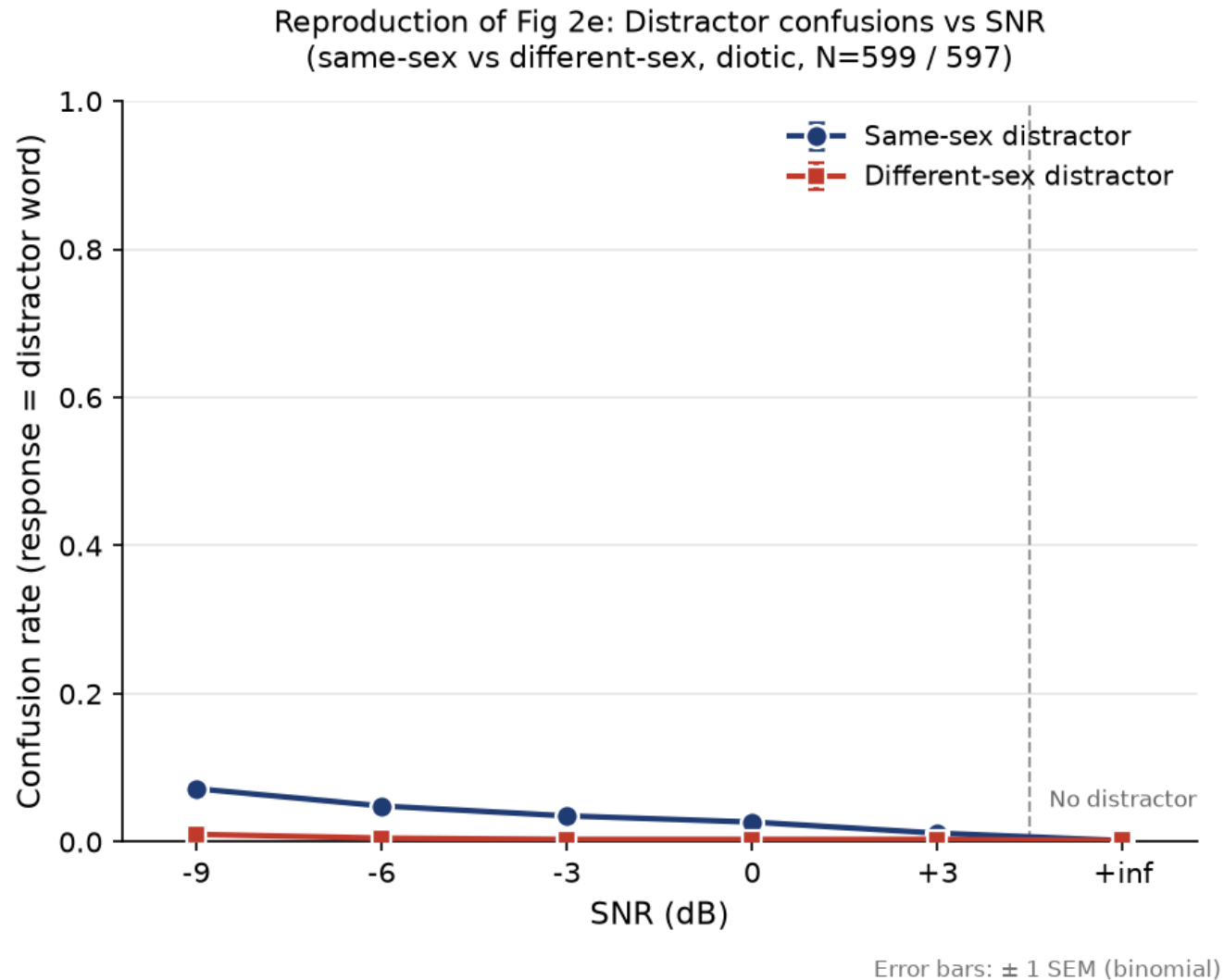
What reproduces

- Monotonic rise with SNR (both lines).
- Different-sex > same-sex at every masked SNR.
- Gap = +9 to +18 percentage points.
- Lines converge to ~88% at +inf — no distractor, sex no longer matters.

within-item, N=599/597, diotic

= paper Fig 2a (same-sex line) + Fig 2b (both lines)

Confusion trend reproduces — but magnitude is too low



Confusion = model reports a distractor word

Direction correct:

- lower SNR $\rightarrow$ more confusion
- same-sex confuses more than different-sex

But 4–5x too low vs the paper

Root cause: distractors were picked at random, so their middle word is usually NOT in the vocabulary — the model cannot output it even if it hears it.

$\rightarrow$ **This is exactly what Part 2 fixes.**

PART 2 · NEW

Number of distractors · 1 / 2 / 4 talkers + babble

Does talker NUMBER matter, at fixed masker energy?

Human hearing: more competing talkers → harder, even when total masker energy is held constant.

That effect is **information masking** — being confused by other voices, not simply drowned out by energy.

So the design must hold total masker energy CONSTANT while varying only the number of voices.

Prediction under test: if the feature-gain model captures information masking, accuracy should fall as talkers are added — at matched energy.

N=1 uses the same task and mixing logic as Part 1 — a built-in directional check.

Conditions & nested sampling

Conditions

no-distractor · 1 · 2 · 4 · 8-talker babble · + ISSN control

SNR: -9...+3 dB for maskers; no-distractor is a single +inf point.

Within-item

The SAME 600 targets are re-mixed in every condition × SNR
→ differences come from the condition, not the sample.

Nested sampling

Each target draws a fixed set $d_1 \dots d_8$; two = d_1, d_2 ; four = $d_1 \dots d_4$; babble = $d_1 \dots d_8$.

→ 'two-talker' is literally 'one-talker plus one more person'. Only the count changes.

Anchored distractor pool

Every distractor recording also carries an in-vocab anchor word, sliced like the target.

- 1,200 recordings / 1,200 speakers
- male 600 / female 600
- 0 overlap with target speakers
- 534 distinct anchor words

Now the model CAN output a distractor word → confusion becomes measurable.

Target-to-total-masker SNR isolates the variable

- Each distractor is RMS-aligned, then SUMMED into one total masker.
- The whole masker is scaled so $10 \cdot \log_{10}(P_{\text{target}} / P_{\text{masker}}) = \text{SNR}$; mixture = target + masker.
- $\rightarrow$ 1 / 2 / 4-talker have **identical total masker energy** at a given SNR.
- So any accuracy difference between them CANNOT be loudness — it is 'more confusable voices'.
- This matches the paper's core 2-/4-talker construction: equalize sources, sum, then set target-to-total-masker SNR.
- The 8-talker babble remains an approximation: ours is summed online; the paper used pre-made files.

Two safeguards added before running

Gender balancing — a real confound

Random draws made $P(\geq 1 \text{ same-sex distractor})$ climb with N: 50% → 99.6%.

Sex is a LARGE effect (Part 1) — it would masquerade as a talker-number effect.

Fix: alternate d1...d8 by sex → same-sex count exactly 0.5 / 1 / 2 / 4.

Verified across all 600 targets: zero jitter, nesting intact, no self / duplicate speakers.

ISSN control condition

Spectrally-matched STATIONARY noise — gapless and contentless.

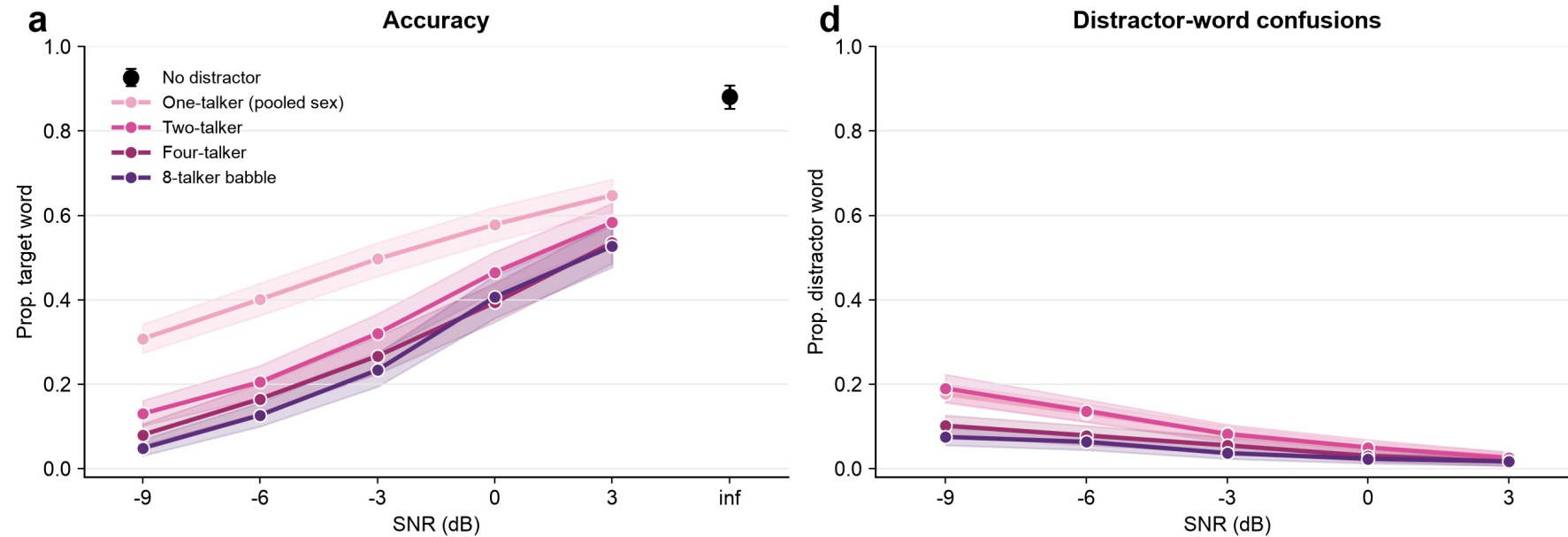
Separates information masking from 'glimpsing' into gaps between words.

If glimpsing drove the effect, speech (which has gaps) would be EASIER than gapless noise.

Also: 1-talker split into one_same / one_diff → free Part-1 (Fig 2a/2b) anchors.

More competing talkers reduce target-word accuracy

Experiment 1: Number of distractors



Single pretrained checkpoint · N = 600 trials / 149 target speakers · 95% cluster-bootstrap CI · One-talker pools same- and different-sex trials

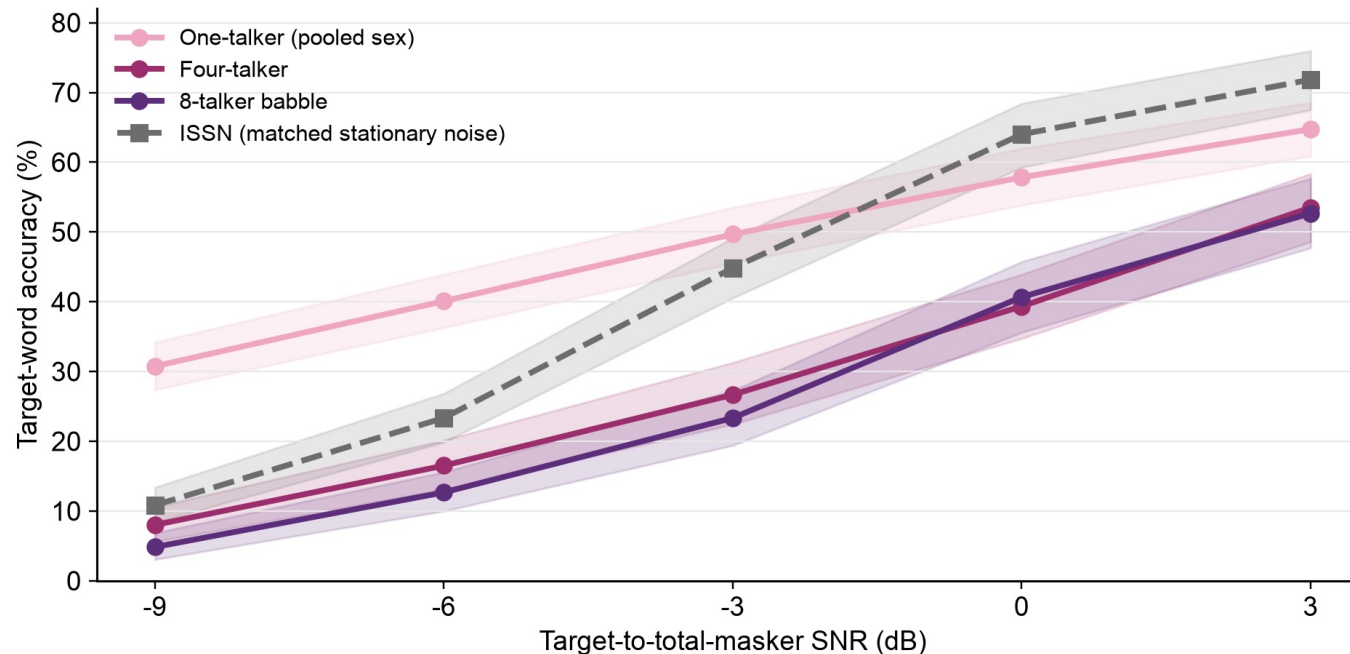
Accuracy (left): monotonic in SNR; pooled one > two > four ≈ babble; +inf = 88%. **Confusion (right):** specific distractor-word reports fall as the masker becomes more noise-like.

Paper-compatible scoring · pooled one-talker sex · 95% CI clustered by 149 target speakers

Speech maskers remain harder than matched stationary noise

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Pattern is consistent with information masking beyond temporal glimpsing

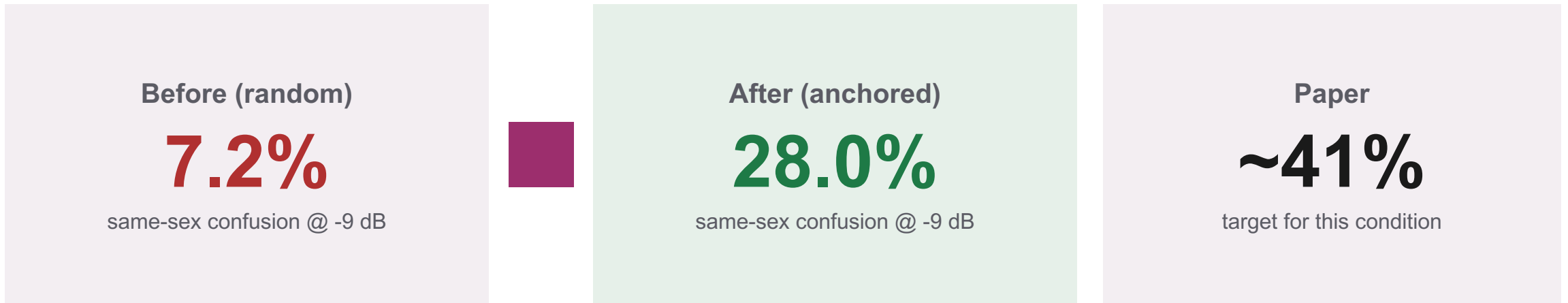


Paper-compatible target scoring · 95% cluster-bootstrap CI by target speaker

ISSN = spectrally-matched stationary noise (gapless, no words)

- Four-talker and babble accuracy is below ISSN at every tested SNR.
- ISSN is gapless, whereas speech contains pauses that should permit glimpsing.
- The remaining speech disadvantage is consistent with information masking beyond gap-listening.
- This is supportive evidence, not a proof: corpus and masker construction still differ from the paper.

The anchored pool fixed the confusion underestimate



From 5.7x too low to 1.5x too low — the root-cause diagnosis was correct.

An honest reversal: confusion is HIGHEST for a single same-sex talker and DROPS as talkers are added — opposite to our pre-registered guess, but consistent with the paper. With many voices the masker becomes noise-like, so the model emits garbage rather than a specific distractor word.

Self-consistency checks — why the pattern is credible

- **no-distractor = 88.0%** — matches the +inf anchor from Part 1 exactly.
 - **one_same / one_diff track the Part-1 single-distractor curves**, 2–6 points lower (expected: a different, anchored distractor pool).
 - **N=1 reduces to the single-distractor case** that was already reproduced — direction and shape hold.
 - **Gender balancing verified across all 600 targets**: exact same-sex counts, nesting intact, no self/duplicate speakers.
- **The results are internally coherent; the exact Windows run script still needs to be archived locally.**

SUMMARY

What reproduced



Model validity (clean)

88.6% vs 0.1% chance



Accuracy vs SNR

monotonic rise, both experiments



Different-sex > same-sex

+9 to +18 percentage points



More talkers → harder

pooled one > two > four ≈ babble



Confusion direction

lower SNR & same-sex → more



Confusion magnitude

fixed 7.2% → 28% (paper ~41%)

Honest boundaries

Corpus mismatch (CV $\neq$ SWC) — the biggest one:

- our stimuli are Common Voice; the paper's Experiment 1 uses held-out Spoken Wikipedia (SWC).
- 231/347 unique target files and 703/1,200 pool files belong to the CV training split.
- split membership does not prove every clip was sampled during training, but held-out generalization cannot be claimed.
- the within-panel contrast is informative, but absolute magnitudes and generalization remain uncertain.
- Confusion vs talker number came out opposite to our stated criterion (reported, not hidden).
- babble = 8 talkers summed online (paper uses pre-made files) — a deliberate, nesting-preserving choice.
- One checkpoint only; the paper's feature-gain curve averages ten architectures.
- Qualitative trend replication only; diotic and custom-stimulus conditions.

Next steps

- Archive the exact RTX 3070 run script, environment, random seed and file hashes.
- Rebuild the evaluation set from held-out Spoken Wikipedia for a paper-faithful replication.
- Use the paper's pre-made babble and exact 2-/4-talker sex composition.
- Run all ten feature-gain architectures, or explicitly retain the single-checkpoint scope.
- Fit a mixed-effects model / speaker-cluster bootstrap and compare directly with OSF curves.

A single frozen feature-gain model qualitatively reproduced the SNR, distractor-sex and talker-number trends.

Thank you — questions?